

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking data

CIM code	MITO-3
Program name	Industrial Technology and Management (INTM)
year	2024-25
Program assessment coordinator	James Mann
Assessment plan year	2024
No assessment this year per plan	☐
LO(s) assessed	LO4 and LO5
Presented at program authority meeting date: <i>Please attach minutes</i>	3/11/2025, 3/12/2025, 4/11/2025, 10/01/2025

Methodology

Three of the MITO (Master's Industrial Technology and Operations) program courses were selected from a previously developed curriculum assessment map, and faculty collected data to assess program learning outcomes (LO) LO#4 and LO#5 for the AY2024-25 period. The INTM curriculum is delivered by Adjunct faculty. One marked change for AY2024-25 is that Adjunct faculty of record were formally engaged and participated in the program assessment process. This is noted in the final discussion. **Table 1** outlines the courses selected and related LOs.

Table 1. INTM MITO graduate courses, learning objectives, and artifacts for the 2024-25 assessment

			LO#4: Identify how current and/or emerging technologies are used to facilitate operational improvements	LO#5: Identify activities which contribute to achieving sustainable industrial operations (sustainable refers to supporting industrial operations over time)
INTM Course	Description	Instructor	Artifact	Artifact
406*	Quality Management Systems	Rama Shankar	Midterm exam Quiz 2	Not evaluated
530	Supply Chain Management	Theo Rigopoulos	Writing Assignment 1 and 2 Midterm exam	Writing Assignment 1 and 2 Midterm exam
561	Energy Options for Industry	Mark Barry	Final exam essay question # 51 / #56	Final exam essay question # 51 / #56

*Note INTM 406 is eligible for undergraduate and graduate curriculum credit

Assessment rubric

The faculty of record were instructed to provide at least two sample artifacts to support each LO. The INTM curriculum graduate and undergraduate courses are generally taught in the same section, with the graduate level course adding structural requirements. The faculty were asked to provide an assignment or exam that best demonstrates each of learning objectives (one sample artifact can apply to both learning objectives). The faculty were instructed that the artifacts need to demonstrate evaluated student work (e.g. a graded assignment).

The artifacts were assembled by the faculty of record based on the faculty interpretation of the learning objectives LO#4 and LO#5 and the assessments were completed with respect to the learning objective rubric. The rubric placed students in three (3) assessment categories: **D** *Does not meet expectations*, **M** *Meets expectations*, and **P** *Proficient*

Table 2. AY 2024-25 curriculum assessment rubric

Learning Objective	D (Does not meet expectations)	M (Meets expectations)	P (Proficient)
LO# 4 Identify how current and/or emerging technologies are used to facilitate operational improvements	- The artifact fails to identify a technology (or it is not clearly related to a technology) and/or the operational improvement is not clear - Technology and improvement are identified but the process of how improvements are applied is not addressed	- At least one technology is clearly identified AND - At least one operational improvement is clearly identified AND - The relationship between the technology and the operational improvement is evident (how they are related)	- <u>Multiple</u> technologies and/or <u>multiple</u> improvements are clearly identified OR - Suggestions on how continuous improvement could be accomplished in the context of these technologies or operational improvements OR - Technologies or combinations of technologies and approaches that are outside state-of-the art and/or creative/innovative solutions identified to apply them
LO#5 Identify activities which contribute to achieving sustainable* industrial operations	- None or very weak connection to sustainability* and/or industrial operations - Artifact fails to communicate clearly a contribution to achieving sustainable* operations (example writing is unclear or visual context is confusing)	- The artifact conveys clearly how the activity(ies) is (are) related to achieving sustainable* industrial operations AND - The artifact demonstrates a clear comprehension of sustainability*	- The artifact includes additional details and/or visual content that describes an impact on sustainability* in industrial operations OR - Best practices and innovative solutions beyond conventional approaches to sustainability *

Results

Artifact samples

Artifact data sets were provided by the faculty of record for each LO#3 and LO#6 in the INTM undergraduate courses. Course enrollments were generally <20 students. The artifact samples are stored by course number in a separate PDF file on a MS One Drive administered by the INTM Staff.

Data table

For each LO the data tables are summarized below. Personally identifiable information is omitted. The summary graphs show the assessment distribution. In comparison to prior assessments, these data tables were prepared by the individual faculty of record.

INTM LO#4	Artifact Description/ Quiz 2		
student	major	D/M/P	comment for N (or other)
student 1	INTM406	P	
student 2	INTM406	P	
student 3	INTM406	P	
student 4	INTM406	P	
student 5	INTM406	P	
student 6	INTM406	P	
student 7	INTM406	P	
student 8	INTM406	P	
student 9	INTM406	P	
student 10	INTM406	P	
student 11	INTM406	P	
student 12	INTM406	P	
student 13	INTM406	P	
student 14	INTM406	P	
student 15	INTM406	P	
student 16	INTM406	N	Withdrawn after midterm
student 17	INTM406	M	

INTM406

Category	Count
D	0
M	1
P	15
N	1

INTM406

Category	Percentage
D	0%
M	5%
P	85%
N	10%

INTM 406 Data table and summary graph

INTM LO#4 AND 5		My Dinner Table Supply Chain				
student	major	D/M/P	comment for N (or other)			
student 1	INTM	P				
student 2	INTM	M				
student 3	INTM	P				
student 4	INTM	D				
student 5	INTM	M				
student 6	INTM	M				
student 7	INTM	P				
student 8	INTM	M				
student 9	INTM	M				
student 10	INTM	P				
student 11	INTM	P				
student 12	INTM	P				
student 13	INTM	P				
student 14	INTM	P				
student 15	INTM	P				
student 16	INTM	P				
student 17	INTM	P				
student 18	INTM	M				
student 19	INTM	P				
student 20	INTM	P				
student 21	INTM	P				

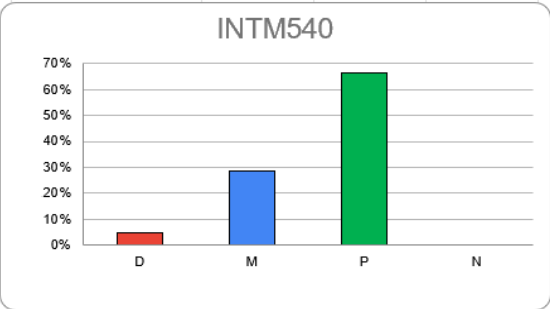
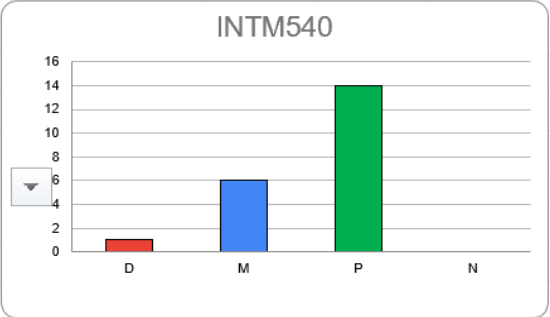


Table 1. INTM 540 Data table and summary graph – Artifact 1

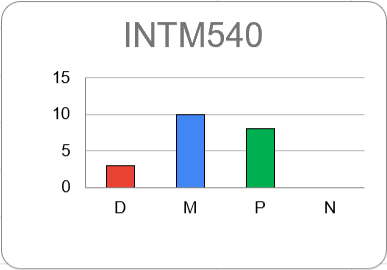
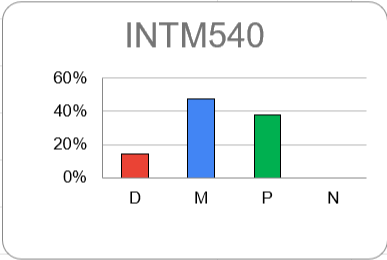
INTM LO#4 AND 5		ATW CONTRACTS		
student	major	D/M/P	comment for N (or other)	
student 1	INTM	M		
student 2	INTM	M		
student 3	INTM	M		
student 4	INTM	D		
student 5	INTM	D		
student 6	INTM	M		
student 7	INTM	M		
student 8	INTM	P		
student 9	INTM	P		
student 10	INTM	P		
student 11	INTM	P		
student 12	INTM	M		
student 13	INTM	M		
student 14	INTM	M		
student 15	INTM	M		
student 16	INTM	P		
student 17	INTM	P		
student 18	INTM	P		
student 19	INTM	D		
student 20	INTM	P		
student 21	INTM	M		

Table 2. INTM 540 Data table and summary graph – Artifact 2

INTM LO#4 AND 5		MIDTERM EXAM
student	major	D/M/P
student 1	INTM	P
student 2	INTM	P
student 3	INTM	M
student 4	INTM	M
student 5	INTM	P
student 6	INTM	P
student 7	INTM	P
student 8	INTM	D
student 9	INTM	P
student 10	INTM	P
student 11	INTM	M
student 12	INTM	D
student 13	INTM	M
student 14	INTM	D
student 15	INTM	M
student 16	INTM	P
student 17	INTM	M
student 18	INTM	M
student 19	INTM	D
student 20	INTM	M
student 21	INTM	M

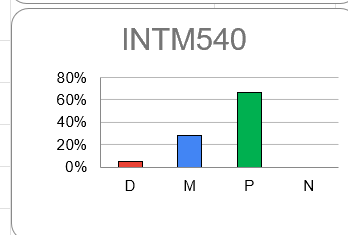
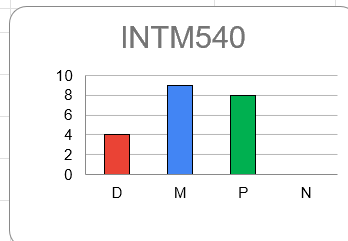


Table 3. INTM 540 Data table and summary graph – Artifact 3

[illegible]

Description of findings

To address this, current plans involve conducting a post-review session with the course faculty. This debrief will focus on analyzing student performance data to uncover possible correlations between low scores and specific content areas tied to the assessed learning outcomes. For instance, if the rubric evaluates outcomes like "apply industrial management principles to real-world scenarios," the review might reveal issues such as insufficient emphasis on certain topics (e.g., supply chain optimization) or challenges in remote learning environments. Faculty could then recommend targeted interventions, such as supplemental resources, revised assignment prompts, or additional in-class examples, to bolster student success in future iterations.

Continuous Improvement, CI, plans

The INTM program assessment strategy is centered on a commitment to continuous improvement, which emphasizes iterative enhancements based on evidence. The goal is to establish a single, streamlined process that integrates three key elements:

Defining Expected Learning Outcomes at the Course Level: This involves clearly articulating what students should know and be able to do by the end of each course. For example, outcomes might include "demonstrate proficiency in data-driven decision-making" or "analyze ethical considerations in technology management." These should be specific, measurable, achievable, relevant, and time-bound (SMART) to facilitate effective evaluation.

Administering a Grading Rubric to Measure Performance: A standardized rubric will be used during the course to assess what students have actually learned. This tool provides objective criteria (e.g., scales from 1-4 for dimensions like content knowledge, application, and communication) and ensures consistency across instructors, including adjuncts. It shifts focus from mere grades to demonstrated competencies.

Post-Hoc Assessment via Audits: After course completion, audits will evaluate the extent to which student learning aligned with expectations at multiple levels—the individual course, the broader INTM/Armour College level, and the institutional Illinois Tech level. This could involve sampling artifacts, analyzing aggregate data, and comparing against benchmarks to identify systemic trends.

This unified process aims to create a feedback loop that is efficient, scalable, and less burdensome for faculty, ultimately fostering a culture of data-informed decision-making.

Prior CI plan impact

Previous CI initiatives have already yielded tangible benefits, particularly in adjunct faculty involvement. A key prior plan identified the need for greater participation from adjuncts of record (faculty teaching specific courses) in the assessment process. Implementation involved providing training on assessment tools, such as spreadsheet-based data entry and artifact submission to a centralized database. Despite some initial hurdles (e.g., formatting issues in spreadsheets or delays in data transfer), the results were positive: adjuncts successfully contributed, leading to more representative data and confirming their capability with proper guidance.

As a direct outcome, engaging adjunct faculty has become a standard operating procedure for the INTM program. This ensures consistency across semesters, leverages practical insights to refine assessments, and builds a more inclusive faculty community. For instance, adjuncts might now routinely review rubrics before teaching, ensuring assignments align with program goals and reducing discrepancies in evaluation.

This cycle's CI plan

Building on the successes and lessons from the AY24-25 assessment cycle, the following CI plan outlines actionable steps to further strengthen the program. These recommendations emphasize sustainability, communication, and collaboration:

Continue Engaging Adjunct Faculty in the Assessment Process: Maintain and expand this SOP by scheduling regular check-ins and providing ongoing support, such as templates for artifact collection. This ensures adjuncts remain integral, potentially increasing assessment completion rates and data quality.

Improve the MS PowerPoint Presentation for Program Assessment: Update the existing presentation used for training and communication. Enhancements could include clearer visuals on rubric application, step-by-step guides for artifact submission, and examples of common pitfalls (e.g., data formatting errors). This will make onboarding adjuncts more efficient and reduce communication challenges.

Adjust Course Syllabi to Refine Alignment: Review and revise syllabi to better connect course objectives with student outcomes (measurable results) and overarching program learning objectives. This alignment ensures assessments directly support INTM program goals.

Identify a Process to Include Adjunct Faculty in the Future Assessment Committee: Develop a formal mechanism, such as rotating adjunct seats on the committee or virtual participation options. This will increase perspectives, foster ownership, and help tailor assessments to adjunct-taught courses, which are typically comprised of evening or online sections.

Propose Iterative Improvements to Course Syllabi and Artifact Collection: Establish an annual review cycle for syllabi updates and artifact gathering. This could involve piloting new artifact types that better align with learning objectives, followed by evaluations of their effectiveness. Iterative testing ensures ongoing relevance, such as incorporating emerging topics like AI in industrial management.

By implementing these elements, the INTM program positions itself for sustainable delivery, with regular reviews to track progress and adapt as needed. This holistic approach not only meets HLC requirements but also drives meaningful enhancements in teaching and learning.

Minutes from meetings for INTM Program Assessment

Meeting Dates: 3/11/2025, 3/12/2025, 4/07/2025

Committee Meeting

3/11/2025, 3/12/2025 James Mann (Director/Industry Professor, INTM), Nick Villani (Assistant Industry Professor, INTM), Mark Barry (Adjunct Faculty, INTM)

An improvement over past assessments the program assessment committee for INTM had three separate meetings in Spring 2025 to prepare a rubric and provide it to faculty for the Spring and Fall assessment.

The INTM faculty are more informed now about the process. As stated in prior assessments the INTM faculty have not previously been charged with assessment related efforts. Going forward all the faculty (including Adjunct) need to be incorporating the learning objectives in the curriculum and completing the assessment for their courses each semester. The faculty were provided the rubric for course level learning objectives in advance during this AY assessment.

Faculty alignment of grading rubrics and the learning outcomes assessment improve overall performance.

Require faculty to explicitly align the course learning objectives / outcomes, as stated in the syllabus, with HLC learning objectives / outcomes. The course grading rubric should be aligned similarly with the HLC assessment rubric. Ultimately, student course grades would measure and reflect how well each student met expectations, i.e. how much they learned.

Faculty Meeting

04/07/2025 James Mann, Brian O'Kane, Rama Shankar, Theo Rigopoulos, Mark Barry, Rama Shankar

The INTM Director, James Mann, communicated the expectations to all Adjunct faculty that they would be formally responsible for collecting artifacts and completing a learning objective assessment using a rubric developed by the INTM Assessment Committee. The INTM course assessment procedure was reviewed, including the specific learning objectives for AY2024-25. The sample spreadsheet for assessment summary was reviewed with the Adjunct faculty and subsequently e-mailed to all faculty. A MS OneDrive folder was created and shared with Adjunct faculty for a repository of the artifacts.

Minutes from meetings for INTN Program Assessment

Meeting Dates: 3/11/2025, 3/12/2025, 4/07/2025

Committee Meeting

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